

ORIGINAL ARTICLE

Proteomics Profiling Reveals Molecular Subtypes of Transthyretin Cardiac Amyloidosis

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BACKGROUND: Transthyretin cardiac amyloidosis (ATTR-CA) may involve heterogeneity in biological mechanisms (ie, molecular subtypes). Molecular subtypes potentially lead to differences in prognosis and treatment response to tafamidis—the first disease-modifying therapy for ATTR-CA. We aimed to derive molecular subtypes of ATTR-CA through comprehensive plasma proteomics profiling and to examine their associations with prognosis and estimated treatment effect of tafamidis.

METHODS: We applied unsupervised machine learning to comprehensive plasma proteomics profiling of 7289 proteins to identify molecular subtypes in our prospective cohort of patients with ATTR-CA enrolled at Columbia University. We compared all-cause mortality risk and estimated the treatment effect of tafamidis among the identified molecular subtypes, adjusting for age, sex, TTR genotype, and concomitant medication use. We also performed pathway analysis comparing the worst prognosis subtype with the other subtypes combined.

RESULTS: Among 142 patients eligible for analysis, 96 patients received tafamidis treatment. We identified 3 molecular subtypes (subtype A: n=49, subtype B: n=38, and subtype C: n=55) with different survival risks during a median follow-up of 5.2 years (log-rank $P=0.035$), without significant differences in traditional clinical stages. Mortality risk was highest in subtype A, followed by B and then C (eg, adjusted hazard ratio of subtype A versus C [95% CI], 1.95 [1.03–3.66]; $P=0.04$). The largest treatment effect of tafamidis was observed in subtype A compared with no tafamidis use within the same subtype (adjusted hazard ratio for mortality, 0.24 [0.10–0.55]), indicating heterogeneous treatment effects of tafamidis (P for interaction=0.001). Subtype A had dysregulation of Ras/MAPK (mitogen-activated protein kinase) pathways, pathways implicated in ATTR-CA pathogenesis, and metabolic/inflammatory pathways.

CONCLUSIONS: Our prospective cohort study using comprehensive plasma proteomics profiling not only uncovered molecular ATTR-CA subtypes but also identified pathogenetic mechanisms associated with differential prognosis and estimated the treatment effect of tafamidis.

Key Words: cardiomyopathies ■ genotype ■ machine learning ■ plasma ■ proteomics

Transthyretin cardiac amyloidosis (ATTR-CA) is a progressive cardiomyopathy due to transthyretin amyloid deposition in the heart.^{1,2} Approximately 1 in 3 patients with asymptomatic ATTR-CA develop symptoms of heart failure (HF) within 4 years of diagnosis, and almost as many die or require heart transplantation.³ Tafamidis, a transthyretin stabilizer, is the first disease-modifying therapy (DMT) for ATTR-CA and has recently become a standard of care for patients with ATTR-CA.⁴ Moreover, some new transthyretin stabilizers (eg,

acoramidis⁵) and transthyretin silencers (eg, patisiran,⁶ eplontersen,⁷ and vutrisiran⁸) have shown efficacy in ATTR-CA and may become options as DMT to switch from or add to tafamidis. Therefore, understanding prognosis and the estimated treatment effect of tafamidis in patients with ATTR-CA is important for precision medicine, including personalized initiation, switching, or addition of DMT.

The prognosis of patients who are diagnosed with ATTR-CA has been estimated using clinical scoring

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WHAT IS NEW?

- Patients with transthyretin cardiac amyloidosis can be classified into 3 molecular subtypes based on comprehensive plasma proteomics profiling using an unsupervised machine learning approach.
- The identified 3 subtypes showed different survival risks and estimated treatment effects of tafamidis.
- The heterogeneous pathophysiology of transthyretin cardiac amyloidosis can involve dysregulation of the Ras/MAPK (mitogen-activated protein kinase) pathways, pathways implicated in transthyretin cardiac amyloidosis pathogenesis (ie, protein processing in the endoplasmic reticulum, Fc gamma receptor-mediated phagocytosis, and autophagy), and metabolic/inflammatory pathways.

WHAT ARE THE CLINICAL IMPLICATIONS?

- The identified molecular subtypes may help inform future precision-medicine strategies for patients with transthyretin cardiac amyloidosis, including decisions about switching from tafamidis to another disease-modifying therapy or adding another disease-modifying therapy to tafamidis.
- In addition, this study can serve as a first step toward understanding the molecular mechanisms underlying the 3 identified molecular subtypes, their different prognoses, and their different treatment effects of tafamidis.

Nonstandard Abbreviations and Acronyms

ACE	angiotensin-converting enzyme
aHR	adjusted hazard ratio
ATTR-CA	transthyretin cardiac amyloidosis
DMT	disease-modifying therapy
FDR	false discovery rate
HF	heart failure
IL	interleukin
MRA	mineralocorticoid receptor antagonist
t-SNE	t-distributed stochastic neighbor embedding

systems (ie, Mayo stage,⁹ National Amyloidosis Center ATTR stage,¹⁰ and Columbia score¹¹) developed before tafamidis became available. Thus, little is known about the factors that predict prognosis and estimated treatment effect of tafamidis in the era of multiple DMTs.¹² Furthermore, given the potential pathological heterogeneity in ATTR-CA,¹³ molecular-level information may provide us with biological mechanisms (ie, molecular subtypes) associated with prognosis and estimated treatment effect of tafamidis. Indeed, molecular subtyping has been explored to make advances toward personalized medicine in cardiovascular diseases, such as HF

with reduced ejection fraction,¹⁴ HF with preserved ejection fraction,¹⁵ and hypertrophic cardiomyopathy.¹⁶ However, no studies have investigated the prognostication or estimated treatment effect of tafamidis in patients with ATTR-CA by taking the presence of molecular subtypes into consideration.

Therefore, in the present prospective cohort study, we aimed to derive molecular subtypes in patients with ATTR-CA using comprehensive plasma proteomics profiling. We then investigated the association of each molecular subtype with mortality and estimated the treatment effect of tafamidis to elucidate differences among the subtypes.

METHODS

Ethics Statement

This study involves human participants and was approved by the Columbia University Irving Medical Center institutional review board (AAAK6507). Participants gave informed consent to participate in the study before taking part.

Study Design and Sample



The present study was a prospective cohort study of patients with ATTR-CA aged ≥ 18 years. Patients were recruited and enrolled at Columbia University Irving Medical Center (New York) from 2009 to 2023. ATTR-CA was diagnosed by any of the after criteria^{1,2}: (1) transthyretin amyloid deposits on endomyocardial biopsy; (2) transthyretin amyloid deposits confirmed by ^{99m}Tc-pyrophosphate/hydroxymethylene diphosphonate scintigraphy with echocardiographic evidence of left ventricular hypertrophy that was out of proportion to systemic loading conditions (ie, maximum left ventricular wall thickness ≥ 12 mm). The attending cardiologists at the Cardiac Amyloidosis Center of the Columbia University Irving Medical Center confirmed the ATTR-CA diagnosis. They excluded ATTR-CA phenocopies (eg, Fabry disease, Danon disease, and hypertrophic cardiomyopathy) by performing a thorough medical interview and physical examination, and when indicated, additional tests (eg, genetic testing and cardiac magnetic resonance imaging). We collected the baseline characteristics of study participants at the time of enrollment (eg, patient demographics, medical history, laboratory tests, and echocardiographic measurements) in our Cardiac Amyloidosis registry. Written informed consent was obtained from all participants, and the study protocol was approved by the Columbia University Irving Medical Center institutional review board. Data supporting the results of this study are available from the corresponding author on reasonable request.

Blood Sample Processing for Comprehensive Proteomics Profiling

We collected venous blood specimens at the time of recruitment, which generally corresponded to the time of ATTR-CA diagnosis and preceded the initiation of tafamidis. We placed the samples in K₂EDTA-treated tubes and centrifuged for 10 min at 3100 rpm. Immediately after aliquoting, we froze the supernatant plasma at -80°C . We used the SomaScan

aptamer-based proteomics assay (SomaLogic Inc, Boulder, CO) to measure the levels of 7289 plasma proteins in a comprehensive proteomic profile. This assay is highly multiplexed, sensitive, quantitative, and reproducible, and can detect proteins >100 fM with a median coefficient of variation <5% for human proteins. Details of the SomaScan aptamer-based proteomics assay are described in [Supplemental Methods](#). We log-transformed the relative fluorescence units of all plasma protein levels for our analysis.

Molecular Subtype Derivation

We preprocessed the proteomics data by first log₂-transforming the relative fluorescence units and then selecting the 300 proteins with the highest variances. We subsequently conducted standardization of only the selected 300 proteins. To create a distance matrix, we calculated the Euclidean distances. Then, we applied the partitioning around medoids, an unsupervised machine learning algorithm, to the preprocessed proteomics data and identified molecular subtypes based on gap statistics. The gap statistic is a common method to statistically determine the optimal number of clusters in a data set by comparing the observed change in intracluster dispersion to the change expected under the null hypothesis distribution.¹⁷ We visualized the identified molecular subtypes using t-distributed stochastic neighbor embedding, which is a nonlinear dimensionality reduction technique that transforms high-dimensional data into low-dimensional space.¹⁸

Baseline Clinical Characteristics

The baseline characteristics in patients of each molecular subtype were compared using 1-way ANOVA or the Kruskal-Wallis test for continuous variables, and Pearson χ^2 test or Fisher exact test for categorical variables, as appropriate. Continuous variables were represented as mean $\pm$ SD if normally distributed, and as median (25th–75th percentile) if not normally distributed. To investigate whether the year of diagnosis differed across molecular subtypes, we analyzed it as a continuous variable using the Kruskal-Wallis test. Diagnosis year was also divided into 3 clinically meaningful eras based on key milestones in ATTR-CA management: ≤ 2010 (limited awareness and diagnostic capability), 2011 to 2015 (emergence of noninvasive diagnostic algorithms), and ≥ 2016 (approval of tafamidis and widespread adoption of bone scintigraphy).¹

Association of Each Molecular Subtype With Mortality

We investigated the unadjusted longitudinal association of each molecular subtype with mortality during the follow-up study period using Kaplan-Meier survival curves with the log-rank test. Next, we compared the risk of mortality in each subtype using a Cox proportional hazards model adjusting for age, sex, transthyretin genotype (ie, wild type, V122I, and others/unknown), and medication use (ie, tafamidis, loop diuretic, β -blocker, ACE [angiotensin-converting enzyme] inhibitor or angiotensin II receptor blocker, and mineralocorticoid receptor antagonist). We considered the molecular subtype with the lowest risk as the reference group in the Cox proportional hazards model. We performed a sensitivity analysis that included race/

ethnicity and left ventricular EF as additional potential confounders in the original Cox proportional hazards model.

We constructed a simplified causal network to illustrate the assumed causal structure underlying our Cox model ([Figure S1](#)). In this framework, the exposure is the molecular subtype derived from baseline proteomics profiling, and the outcome is all-cause death. Age, sex, transthyretin genotype, and medication use were included as confounders, all of which are preexisting or baseline characteristics determined before proteomics profiling was performed and therefore cannot be mediators or colliders in our causal model. To quantify the robustness of our findings to potential unmeasured confounding, we calculated the *E* value.¹⁹

Association of Each Molecular Subtype With Estimated Treatment Effect of Tafamidis

To examine the treatment effect of tafamidis on survival in each molecular subtype, we developed a Cox proportional hazards model for each subtype, adjusting for age, sex, transthyretin genotype, and medication use (ie, loop diuretic, β -blocker, ACE inhibitor or angiotensin II receptor blocker, and mineralocorticoid receptor antagonist). We examined heterogeneity in the estimated treatment effect of tafamidis in each subtype by calculating the interaction between molecular subtype and tafamidis use.

Pathway Analysis

We identified a molecular subtype with the highest risk of mortality (ie, subsequently identified as subtype A). Next, we used the Mann-Whitney *U* test (Wilcoxon test to determine whether there were statistically significant differences in plasma protein levels between the highest-risk molecular subtype (ie, subtype A) and all the other subtypes combined (ie, subtypes B and C). Finally, a total of 1566 proteins with Benjamini-Hochberg adjusted $P < 0.05$ and log₂ (fold change) >1.2 were included in pathway analysis.

We performed pathway analysis using STRING V.12 to reveal canonical pathways that are differentially dysregulated between the subtype with the worst prognosis and the other subtypes combined.²⁰ We calculated the ratio of proteins mapped to a particular pathway to the total number of proteins associated with that pathway.²⁰ All potential canonical pathways were based on the Kyoto Encyclopedia of Genes and Genomes database.²¹ We reported the biologically relevant pathways based on false discovery rate (FDR; ie, $FDR < 1.0 \times 10^{-5}$) among all differentially dysregulated candidate pathways.

We considered a *P* value of <0.05 as statistically significant and performed all analyses with R (version 4.3.1), unless otherwise indicated.

RESULTS

We enrolled 142 patients with ATTR-CA in the study from our prospective cohort. During a median follow-up of 5.2 years (25th–75th percentile, 3.1–6.5 years) in the 142 patients, 96 (68%) patients were treated with tafamidis. During the follow-up period, a total of 63 deaths (44%) were observed, including 31 (63%)

in subtype A, 15 (39%) in subtype B, and 17 (31%) in subtype C. The median year of diagnosis differed slightly across molecular subtypes: 2016, 2018, and 2017 for subtypes A, B, and C, respectively (Kruskal-Wallis $P<0.001$; Figure S2). Three molecular subtypes were derived according to the gap statistics (Figure S3). The identified molecular subtypes are shown in the t-distributed stochastic neighbor embedding visualization (Figure 1).

Baseline Clinical Characteristics

We delineated the baseline characteristics of the study cohort by the 3 molecular subtypes—molecular subtype A, B, and C (Table 1). We found no statistical differences in patient demographics, transthyretin genotype, vital signs, medical history, laboratory tests, echocardiographic measurements, or ATTR-CA clinical risk stages, indicating that the molecular subtypes cannot be identified without proteomics profiling. Of note, while patients in molecular subtype A were less likely to receive tafamidis, patients with molecular subtype C were less likely to receive loop diuretics, ACE inhibitors or angiotensin II receptor blockers, and mineralocorticoid receptor antagonists.

Association of Each Molecular Subtype With Prognosis

The risks of mortality were significantly different among the 3 molecular subtypes (log-rank test, $P=0.035$), as shown in Figure 2. The proportional hazards assumption was confirmed by the global Schoenfeld residuals test ($P=0.43$). Patients in molecular subtype A had the highest risk of mortality among all the groups. Molecular subtype C had the lowest risk and was treated as the reference group. After adjustment for age, sex, transthyretin

genotype, and medication use in the Cox proportional hazards model, molecular subtype A continued to have a significantly higher risk of death compared with the reference group (adjusted hazard ratio [aHR] of subtype A versus C [95% CI], 1.95 [1.03–3.66], $P=0.04$; Table 2). Consistent results were observed when race/ethnicity and left ventricular EF were added to the original set of potential confounders in the Cox proportional hazards model (Table S1). The E value for the aHR of subtype A versus C (aHR=1.95) was 3.31, indicating that unmeasured confounding, whether due to 1 or multiple unmeasured factors, would need to be associated with both molecular subtype and mortality by a risk ratio of ≥ 3.31 , above and beyond the measured confounders, to fully explain away the observed association.

Association of Each Molecular Subtype With Estimated Treatment Effect of Tafamidis

The largest estimated treatment effect of tafamidis was observed in patients with molecular subtype A (aHR of tafamidis use, 0.24 [95% CI, 0.10–0.55]; Figure 3) compared with subtype B (aHR, 0.42 [95% CI, 0.10–1.79]) and subtype C (aHR, 0.50 [95% CI, 0.15–1.69]). Testing for effect modification between the molecular subtypes and tafamidis use also demonstrated heterogeneous estimated treatment effects of tafamidis (P for interaction=0.001). Events and event rates by molecular subtype and tafamidis use are presented in Table S2, and Kaplan-Meier survival curves stratified by tafamidis use within each molecular subtype are shown in Figure S4.

Pathway Analysis

We plotted the results from pathway analysis in Figure 4. Pathway analysis comparing the subtype with the worst prognosis (ie, molecular subtype A) with the

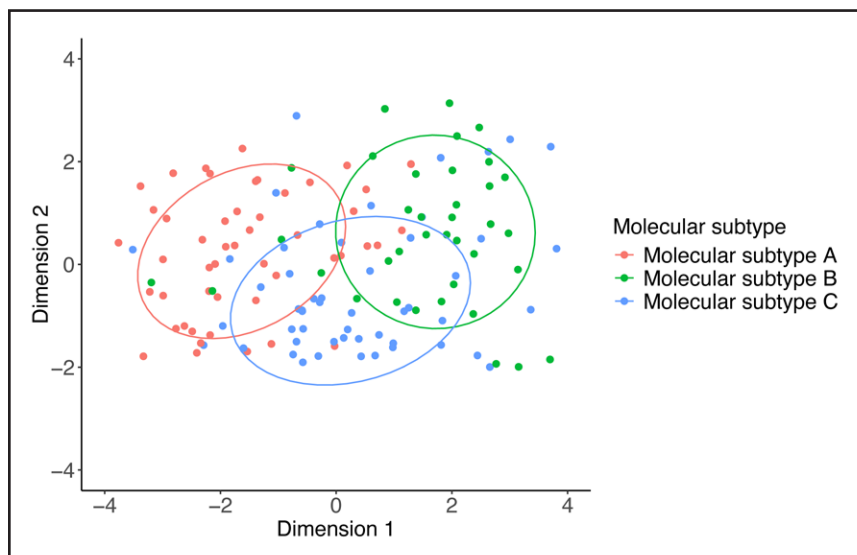


Figure 1. t-distributed stochastic neighbor embedding (t-SNE) visualization of different molecular subtypes.

We derived 3 molecular subtypes by applying unsupervised machine learning to plasma proteomics profiling data in our prospective cohort of 142 patients with transthyretin cardiac amyloidosis. The identified molecular subtypes are shown in the t-SNE visualization. t-SNE is a nonlinear dimensionality reduction technique that transforms high-dimensional data into low-dimensional space for visualization.

Table 1. Baseline Clinical Characteristics of the Study Cohort by Molecular Subtype

Characteristics	Molecular subtype A (n=49)	Molecular subtype B (n=38)	Molecular subtype C (n=55)	P values
Patient demographics				
Age	73±8	76±7	74±9	0.27
Female sex	6 (12)	4 (11)	6 (11)	0.97
Race and ethnicity				0.97
Non-Hispanic White	39 (80)	30 (79)	48 (87)	
Non-Hispanic Black	5 (10)	3 (8)	3 (6)	
Hispanic	2 (4)	1 (3)	1 (2)	
Other/unknown	3 (6)	4 (11)	3 (6)	
Body mass index, kg/m ²	26.3±3.9	26.7±4.3	26.2±3.7	0.86
TTR genotype				0.83
Wild type	37 (76)	32 (84)	41 (75)	
V122I	6 (12)	2 (5)	5 (9)	
Others/unknown	6 (12)	4 (11)	9 (16)	
Vital signs				
Systolic blood pressure, mmHg	119±15	118±18	122±17	0.46
Diastolic blood pressure, mmHg	73±11	69±9	74±10	0.20
Heart rate, beat/min	77±13	74±15	77±16	0.69
Medical history				
NYHA functional class	2 (2–3)	2 (2–3)	2 (2–2)	0.16
Hypertension	10 (20)	12 (32)	19 (36)	0.22
Diabetes	6 (12)	3 (8)	3 (6)	0.49
Atrial fibrillation	27 (55)	23 (61)	33 (62)	0.75
Coronary artery disease	15 (31)	12 (32)	14 (26)	0.84
Stroke	7 (14)	9 (24)	4 (8)	0.10
Chronic kidney disease	22 (46)	21 (55)	27 (52)	0.67
Carpal tunnel syndrome	11 (27)	12 (40)	10 (25)	0.36
Spinal stenosis	6 (15)	5 (17)	5 (13)	0.90
Autonomic neuropathy	3 (8)	2 (7)	3 (7)	0.99
Peripheral neuropathy	7 (18)	6 (19)	6 (14)	0.82
Pacemaker implantation	9 (19)	9 (24)	12 (24)	0.81
Medication				
Loop diuretic	35 (73)	28 (74)	26 (49)	0.02
Furosemide dose, mg/kg	0.29 (0.00–0.50)	0.26 (0.00–0.53)	0.00 (0.00–0.34)	0.04
β-Blocker	24 (50)	19 (50)	24 (45)	0.86
ACE inhibitor or ARB	20 (42)	12 (32)	10 (19)	0.04
Mineralocorticoid receptor antagonist	14 (29)	11 (29)	4 (8)	0.01
Tafamidis	25 (51)	32 (84)	39 (71)	0.003
Laboratory tests				
Creatinine, mg/dL	1.23±0.39	1.22±0.35	1.30±0.50	0.63
eGFR, mL/min per 1.73m ²	65.5±27.0	62.1±19.7	61.0±22.9	0.65
NT-proBNP, pg/mL	2805±3115	2615±2157	2328±1717	0.64
hs-cTnT, ng/mL	43±39	60±123	50±113	0.75
Echocardiographic measurements				
Left ventricular ejection fraction, %	47±15	50±12	53±12	0.13
Left atrial volume, mL	84±22	101±49	91±19	0.48
Interventricular septum thickness, mm	16±4	16±4	16±4	0.78

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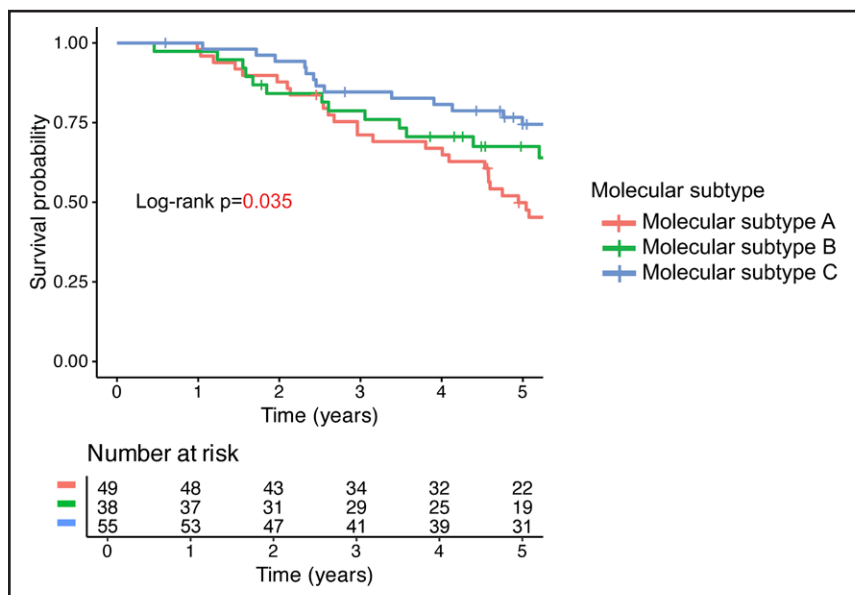
Table 1. Continued

Characteristics	Molecular subtype A (n=49)	Molecular subtype B (n=38)	Molecular subtype C (n=55)	P values
Posterior wall thickness, mm	15±4	15±3	15±4	0.89
Left ventricular end-systolic volume, mL	38±15	42±21	34±14	0.93
Left ventricular end-diastolic volume, mL	91±17	90±30	86±31	0.65
ATTR-CA risk scores				
Mayo stage				0.28
Stage 1	21 (50)	19 (59)	29 (66)	
Stage 2	16 (38)	7 (22)	8 (18)	
Stage 3	5 (12)	6 (19)	7 (16)	
National amyloidosis center stage				0.85
Stage 1	25 (57)	20 (61)	29 (63)	
Stage 2	13 (30)	7 (21)	12 (26)	
Stage 3	6 (14)	6 (18)	5 (11)	
Columbia score stage				0.47
Stage 1	17 (45)	12 (50)	26 (65)	
Stage 2	17 (45)	10 (42)	12 (30)	
Stage 3	4 (11)	2 (8)	2 (5)	
Outcome				
Mortality during the follow-up period	31 (63)	15 (39)	17 (31)	0.003

Values represent mean ± SD, median (25–75 percentile), or n (%). Percentages for categorical variables are based on available data for each variable. Baseline characteristics across the 3 molecular subtypes were compared using 1-way ANOVA or the Kruskal-Wallis test for continuous variables, and the χ^2 test or Fisher exact test for categorical variables, as appropriate. ATTR-CA, transthyretin cardiac amyloidosis; ACE, angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; eGFR, estimated glomerular filtration rate; hs-cTnT, high-sensitivity cardiac troponin T; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; TTR, transthyretin.

other 2 groups combined (ie, molecular subtypes B and C) showed upregulation of Ras/MAPK (mitogen-activated protein kinase)-related signaling pathways, such as the ErbB signaling pathway ($FDR=1.9\times 10^{-9}$), MAPK signaling pathway ($FDR=1.6\times 10^{-8}$), and Ras signaling pathway ($FDR=4.0\times 10^{-7}$), in patients classified as molecular subtype A. In addition, pathways implicated in ATTR-CA pathogenesis, such as protein processing in

the endoplasmic reticulum ($FDR=1.8\times 10^{-8}$), Fc gamma receptor-mediated phagocytosis ($FDR=3.2\times 10^{-7}$), and autophagy ($FDR=2.3\times 10^{-6}$), were differentially dysregulated in patients with molecular subtype A. Pathways involved in metabolism and inflammation were also significantly dysregulated in patients with molecular subtype A (eg, metabolic pathways [$FDR=2.0\times 10^{-12}$], chemokine signaling pathway [$FDR=1.8\times 10^{-9}$]).

**Figure 2. Kaplan-Meier survival curves according to molecular subtypes.**

We compared the Kaplan-Meier survival curves of each molecular subtype. The corresponding adjusted hazard ratios are shown in Table 2. The risks of mortality were significantly different among the 3 molecular subtypes (log-rank $P=0.035$). Molecular subtype A exhibited the highest risk of mortality, while subtype C had the lowest.

Table 2. Association of Molecular Subtype With Death During the Follow-Up Period

Molecular subtype	Adjusted HR (95% CI)	P value
Molecular subtype A	1.95 (1.03–3.66)	0.04
Molecular subtype B	1.37 (0.66–2.86)	0.40
Molecular subtype C	Reference	...

We developed a Cox proportional hazards model to examine the association of molecular subtype with time to death. We adjusted for age, sex, TTR genotype, and medication use (ie, tafamidis, loop diuretic, β -blocker, ACE inhibitor or ARB, and MRA). The corresponding Kaplan-Meier curves are shown in Figure 2. ACE, angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; HR, hazard ratio; MRA, mineralocorticoid receptor antagonist; TTR, transthyretin.

DISCUSSION

Summary of Findings

In the present prospective cohort study of 142 patients with ATTR-CA, we identified 3 molecular subtypes by applying an unsupervised machine learning method to comprehensive plasma proteomics data. One molecular subtype showed not only the highest risk of mortality but also the largest estimated treatment effect of tafamidis, supporting heterogeneity in pathological mechanisms and estimated treatment effects of tafamidis in ATTR-CA. In addition, the pathway analysis revealed that the heterogeneity in ATTR-CA can be explained by the dysregulation of Ras/MAPK pathways, pathways implicated in ATTR-CA pathogenesis, and metabolic/inflammatory pathways.

Results in the Context

Molecular subtyping using proteomics profiling is a useful approach to uncover novel pathological pathways that underlie progression to severe forms of disease and heterogeneity in the treatment effects of specific drugs. For example, Tromp et al¹⁴ reported that a specific molecular subtype in HF with reduced ejection fraction on the basis

of cardiovascular biomarker profiling may benefit less from β -blocker therapy than other molecular subtypes. Woolley et al¹⁵ identified a novel molecular subtype associated with high prognostic risk in patients with HF with preserved ejection fraction using proteomics profiling and found a subtype to have upregulation of inflammatory pathways. Furthermore, our prior study in patients with hypertrophic cardiomyopathy showed that biological differences elucidated by proteomics profiling were associated with higher risk of subsequent major adverse cardiovascular events.¹⁶ These studies collectively underscore the importance of applying molecular subtyping with proteomics profiling to the ATTR-CA population.

Kourelis et al¹³ reported that a molecular subtype characterized by activated PIK3C3, a key protein of the PI3K-Akt pathway, was associated with higher mortality based on selected proteomics profiling of the myocardium in patients with ATTR-CA. However, this study’s generalizability is limited in current clinical practice because ^{99m}Tc scintigraphy has been widely used to diagnose ATTR-CA, and very few patients undergo endomyocardial biopsy, most of whom had an advanced phenotype.^{1,2} Plasma samples would be preferred over myocardial samples for prognostication of patients with ATTR-CA because plasma sampling is less invasive and more easily performed in various clinical settings. In this context, the current study serves as the first to apply comprehensive plasma proteomic profiling to uncover molecular ATTR-CA subtypes and identify underlying pathways associated with different prognoses and estimated treatment effect of tafamidis.

Identified Signaling Pathways and Their Roles in the Pathogenesis of ATTR-CA

The current study showed that the subtype with the highest risk of mortality and the largest estimated treatment effect of tafamidis (ie, molecular subtype

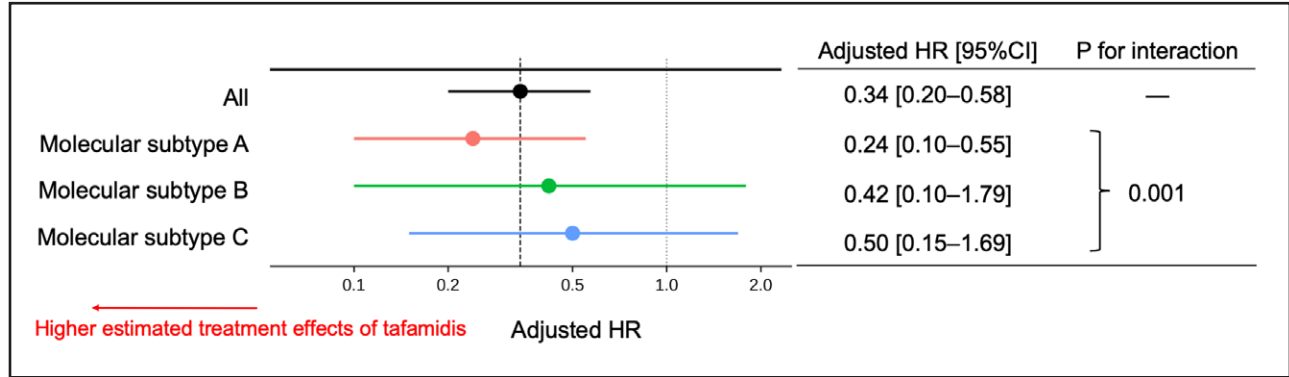


Figure 3. Association of tafamidis use with death in each molecular subtype.

We developed a Cox proportional hazards model in each molecular subtype to examine the association of tafamidis use with time to death and tested for the interaction between molecular subtype and tafamidis use. We adjusted for age, sex, transthyretin genotype, and use of each medication other than tafamidis (ie, loop diuretic, β -blocker, ACE [angiotensin-converting enzyme] inhibitor or angiotensin II receptor blocker [ARB], and mineralocorticoid receptor antagonist [MRA]). Molecular subtype A showed the lowest hazard ratio (HR) for tafamidis use, consistent with heterogeneous estimated treatment effects of tafamidis (P for interaction=0.001).

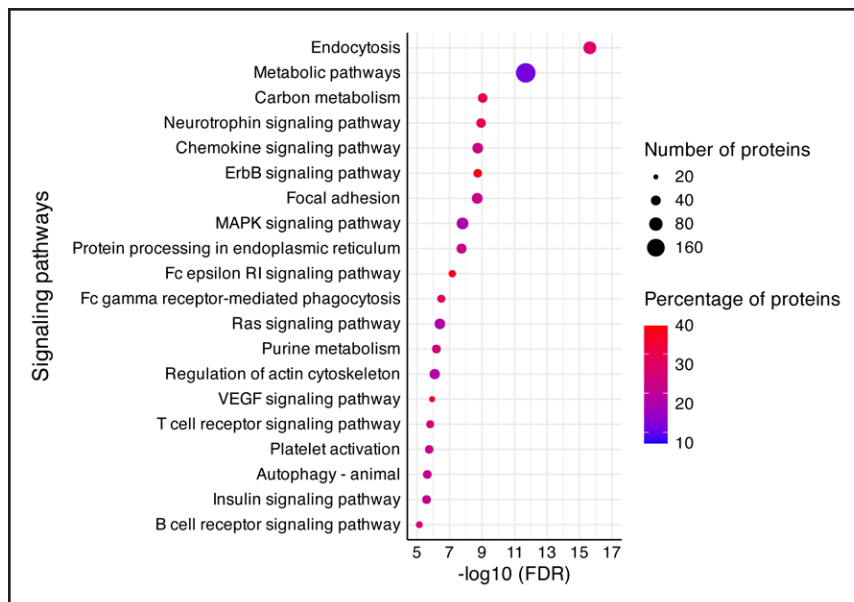


Figure 4. Pathway analysis of proteomics profiling data.

We show differentially regulated signaling pathways (ie, false discovery rate $<1.0 \times 10^{-5}$) in the pathway analysis. We revealed a dysregulation of Ras/MAPK (mitogen-activated protein kinase)-related pathways and pathways implicated in transthyretin cardiac amyloidosis (ATTR-CA) pathogenesis (ie, protein processing in the endoplasmic reticulum, Fc gamma receptor-mediated phagocytosis, and autophagy), as well as metabolic/inflammatory pathways by comparing molecular subtype A with all the other subtypes combined (ie, molecular subtypes B and C). VEGF indicates vascular endothelial growth factor.

A) had dysregulation of Ras/MAPK pathways, pathways implicated in ATTR-CA pathogenesis (ie, protein processing in the endoplasmic reticulum, Fc gamma receptor-mediated phagocytosis, and autophagy), and metabolic/inflammatory pathways. Shi et al²² have shown that amyloidogenic proteins can lead to oxidative stress, cell dysfunction, and apoptosis through activation of p38 MAPK in patients with primary light chain cardiac amyloidosis. Regarding pathways implicated in ATTR-CA pathogenesis, the protein-processing-in-the-endoplasmic-reticulum pathway is critical to control protein folding and stabilize transthyretin tetramers.²³ In addition, while the importance of the Fc gamma receptor-mediated phagocytosis pathway in the context of ATTR-CA has not been extensively reported, the Fc gamma receptor-mediated phagocytosis pathway has been found to be dysregulated in other amyloid disorders, such as Alzheimer disease.²⁴ Autophagy, which plays an important role in removing amyloid²⁵ and reducing proteotoxicity,²⁶ was dysregulated, consistent with the aforementioned study using myocardial proteomics profiling in ATTR-CA.¹³ Indeed, the use of sodium-glucose cotransporter 2 inhibitors, which can promote autophagic flux,²⁷ was suggested to have cardioprotective effects in patients with ATTR-CA.²⁸ Finally, aggregated transthyretin induces the secretion of proinflammatory mediators including IL-6 (interleukin-6) and IL-8,²⁹ which subsequently increases metabolic activity in the heart. Taken together, the signaling pathways that are dysregulated in molecular subtype A suggest that the failure of biological defense mechanisms against transthyretin tetramers and the subsequent intracellular stress response contribute to ATTR-CA disease progression. These findings may partly explain the larger estimated treatment effect of tafamidis in molecular subtype A.

Clinical Implications of Molecular Subtypes in ATTR-CA



This study identified molecular subtypes associated with both prognosis and estimated treatment effect of tafamidis in the era of multiple DMTs. Our approach to identifying ATTR-CA molecular subtypes may help inform future personalized treatment strategies for DMT. For example, if a patient appears to be at high risk of mortality despite tafamidis therapy, physicians may consider switching to other transthyretin stabilizers (eg, acoramidis⁵) or adding transthyretin silencers (eg, patisiran,⁶ eplontersen,⁷ and vutrisiran⁸) to tafamidis. Given that head-to-head clinical trials comparing the estimated treatment effects of each DMT and their combination are unlikely to be performed, the present study suggests that more precise phenotyping using proteomics could help guide clinicians regarding personalized use of DMT for patients with ATTR-CA.

The average estimated treatment effect of tafamidis in our cohort (pooled aHR 0.34) was larger than that reported in ATTR-ACT (HR, 0.70).⁴ One plausible explanation is the earlier detection and treatment afforded by specialized referral centers such as ours. Through proactive screening, our cohort includes a substantial proportion of patients identified before the onset of overt HF, and earlier diagnosis and treatment have been associated with improved outcomes in ATTR-CA.^{30,31}

Strengths of the Present Study

To our knowledge, this is the first study to uncover molecular subtypes and their associated pathogenic pathways in ATTR-CA using plasma proteomics profiling. The SomaScan aptamer-based proteomics profiling we

used in this study is known for its high accuracy, sensitivity, and reproducibility.³² In addition, unsupervised machine learning algorithms allow us to identify potential clusters in a group without assuming hypotheses based on domain knowledge or experience. With these methodological strengths, we successfully identified the molecular subtypes, providing unique and detailed insights into the pathological mechanisms of ATTR-CA that cannot be derived from traditional clinical information. Furthermore, our unique cohort of patients with ATTR-CA with a long follow-up period (ie, median follow-up of 5.2 years) enabled us to demonstrate the association of each molecular subtype with survival and estimated treatment effect of tafamidis. Future identification of ATTR-CA molecular subtypes in the clinical setting by measuring selected proteins using more rapid and inexpensive methods, such as ELISA, could help inform future personalized management strategies with DMT for patients with ATTR-CA.

Potential Limitations

Our study has several potential limitations. First, this was a single-center study conducted in a high-volume tertiary care hospital that serves as a referral center for patients with ATTR-CA; therefore, the findings may have limited generalizability to broader patient populations with ATTR-CA. Second, because we do not have data on each proteomic expression level in the myocardium, we could not confirm that the identified molecular subtypes accurately reflect pathological changes in the myocardium. Third, the current study did not fully address the causality of molecular subtypes and their associated pathways on prognosis and the estimated treatment effect of tafamidis. In particular, the treatment effect estimates from our prospective cohort study may be influenced by confounding by indication and other residual or unmeasured confounders. Fourth, the median year of diagnosis differed by 1 to 2 years among the molecular subtypes. Such a small difference is unlikely to cause prognostic disparities among the subtypes. However, subtype A had the earliest median year of diagnosis and may therefore have included a higher proportion of patients treated with older regimens, potentially affecting their clinical outcomes. Fifth, external validation of the identified molecular subtypes in independent cohorts is needed to confirm the generalizability and reproducibility of our clustering results. Finally, our study used proteomic profiling performed only once at the time of patient enrollment. Future research should explore the ideal frequency, timing, and situation (eg, outpatient versus inpatient settings) for measuring proteomic profiling to optimize plasma biomarker-based molecular subtyping, prognostication of patients with ATTR-CA, and estimated treatment effect.

Conclusions

In the present prospective cohort study of 142 patients with ATTR-CA, we identified 3 plasma proteomics-based molecular subtypes associated with future risk of mortality and estimated treatment effect of tafamidis. Differentially dysregulated pathways were observed in the molecular subtype with the highest risk of mortality and the largest estimated treatment effect of tafamidis. Further elucidation of the molecular mechanisms underlying ATTR-CA disease progression and promotion of a personalized medicine approach for patients with ATTR-CA in the era of multiple DMTs may be facilitated by proteomic profiling. Moreover, our findings may facilitate drug discovery by highlighting novel therapeutic targets and signaling pathways.

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Author Contributions

Drs Osawa and Shimada conceived the research idea. All authors contributed to the data collection. Drs Osawa and Shimada contributed to the design and analysis as well as article preparation. Drs Osawa, Maurer, and Shimada contributed to the interpretation of the data and article revision. Dr Shimada is responsible for the overall content of the article as guarantor. All authors have read and approved the final article.

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Supplemental Material

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REFERENCES

1. Kittleson MM, Maurer MS, Ambardekar AV, Bullock-Palmer RP, Chang PP, Eisen HJ, Nair AP, Nativi-Nicolau J, Ruberg FL; American Heart Association Heart Failure and Transplantation Committee of the Council on Clinical Cardiology. Cardiac amyloidosis: evolving diagnosis and management: a scientific statement from the American Heart Association. *Circulation*. 2020;142:e7–e22. doi: 10.1161/CIR.0000000000000792

2. Garcia-Pavia P, Rapezzi C, Adler Y, Arad M, Basso C, Brucato A, Burazor I, Caforio ALP, Damy T, Eriksson U, et al. Diagnosis and treatment of cardiac amyloidosis: a position statement of the ESC working group on myocardial and pericardial diseases. *Eur Heart J*. 2021;42:1554–1568. doi: 10.1093/eurheartj/ehab072
3. Gonzalez-Lopez E, Escobar-Lopez L, Obici L, Satri G, Bezard M, Saith SE, AbouEzzeddine OF, Mussinelli R, Gagliardi C, Kharoubi M, et al. Prognosis of transthyretin cardiac amyloidosis without heart failure symptoms. *JACC CardioOncol*. 2022;4:442–454. doi: 10.1016/j.jacc.2022.07.007
4. Maurer MS, Schwartz JH, Gundapaneni B, Elliott PM, Merlini G, Waddington-Cruz M, Kristen AV, Grogan M, Witteles R, Damy T, et al; ATTR-CT Study Investigators. Tafamidis treatment for patients with transthyretin amyloid cardiomyopathy. *N Engl J Med*. 2018;379:1007–1016. doi: 10.1056/NEJMoa1805689
5. Gillmore JD, Judge DP, Cappelli F, Fontana M, Garcia-Pavia P, Gibbs S, Grogan M, Hanna M, Hoffman J, Masri A, et al; ATTRIBUTE-CM Investigators. Efficacy and safety of acoramidis in transthyretin amyloid cardiomyopathy. *N Engl J Med*. 2024;390:132–142. doi: 10.1056/NEJMoa2305434
6. Maurer MS, Kale P, Fontana M, Berk JL, Grogan M, Gustafsson F, Hung RR, Gottlieb RL, Damy T, González-Duarte A, et al; APOLLO-B Trial Investigators. Patisiran treatment in patients with transthyretin cardiac amyloidosis. *N Engl J Med*. 2023;389:1553–1565. doi: 10.1056/NEJMoa2300757
7. Masri A, Maurer MS, Claggett BL, Kulac I, Waddington Cruz M, Conceição I, Weiler M, Berk JL, Gertz M, Gillmore JD, et al. Effect of eplontersen on cardiac structure and function in patients with hereditary transthyretin amyloidosis. *J Card Fail*. 2024;30:973–980. doi: 10.1016/j.cardfail.2023.11.016
8. *HELIOS-B: A study to evaluate vutrisiran in patients with transthyretin amyloidosis with cardiomyopathy*. National Library of Medicine, ClinicalTrials.gov. 2024. Accessed June 20, 2026. <https://clinicaltrials.gov/study/NCT04153149>
9. Grogan M, Scott CG, Kyle RA, Zeldenrust SR, Gertz MA, Lin G, Klarich KW, Miller WL, Maleszewski JJ, Dispenzieri A. Natural history of wild-type transthyretin cardiac amyloidosis and risk stratification using a novel staging system. *J Am Coll Cardiol*. 2016;68:1014–1020. doi: 10.1016/j.jacc.2016.06.033
10. Gillmore JD, Damy T, Fontana M, Hutchinson M, Lachmann HJ, Martinez-Naharro A, Quarta CC, Rezk T, Whelan CJ, Gonzalez-Lopez E, et al. A new staging system for cardiac transthyretin amyloidosis. *Eur Heart J*. 2018;39:2799–2806. doi: 10.1093/eurheartj/ehx589
11. Cheng RK, Levy WC, Vasbinder A, Teruya S, De Los Santos J, Leedy D, Maurer MS. Diuretic dose and NYHA functional class are independent predictors of mortality in patients with transthyretin cardiac amyloidosis. *JACC CardioOncol*. 2020;2:414–424. doi: 10.1016/j.jacc.2020.06.007
12. Rapezzi C, Elliott P, Damy T, Nativi-Nicolau J, Berk JL, Velazquez EJ, Boman K, Gundapaneni B, Patterson TA, Schwartz JH, et al. Efficacy of tafamidis in patients with hereditary and wild-type transthyretin amyloid cardiomyopathy: further analyses from ATTR-CT. *JACC Heart Fail*. 2021;9:115–123. doi: 10.1016/j.jchf.2020.09.011
13. Kourelis TV, Dasari SS, Dispenzieri A, Maleszewski JJ, Redfield MM, Fayyaz AU, Grogan M, Ramirez-Alvarado M, Abou Ezzeddine OF, McPhail ED. A proteomic atlas of cardiac amyloid plaques. *JACC CardioOncol*. 2020;2:632–643. doi: 10.1016/j.jacc.2020.08.013
14. Tromp J, Ouwerkerk W, Demissei BG, Anker SD, Cleland JG, Dickstein K, Filippatos G, van der Harst P, Hillege HL, Lang CC, et al. Novel endotypes in heart failure: effects on guideline-directed medical therapy. *Eur Heart J*. 2018;39:4269–4276. doi: 10.1093/eurheartj/ehy172
15. Woolley RJ, Ceelen D, Ouwerkerk W, Tromp J, Figarska SM, Anker SD, Dickstein K, Filippatos G, Zannad F, Metra M, et al. Machine learning based on biomarker profiles identifies distinct subgroups of heart failure with preserved ejection fraction. *Eur J Heart Fail*. 2021;23:983–991. doi: 10.1002/ehfj.2144
16. Liang LW, Raita Y, Hasegawa K, Fifer MA, Maurer MS, Reilly MP, Shimada YJ. Proteomics profiling reveals a distinct high-risk molecular subtype of hypertrophic cardiomyopathy. *Heart*. 2022;108:1807–1814. doi: 10.1136/heartjnl-2021-320729
17. Tibshirani R, Walther G, Hastie T. Estimating the number of clusters in a data set via the gap statistic. *J R Stat Soc Ser B Stat Methodol*. 2001;63:411–423. doi: 10.1111/1467-9868.00293
18. van der Maaten L, Hinton G. Visualizing data using t-SNE. *J Mach Learn Res*. 2008;9:2579–2605
19. VanderWeele TJ, Ding P. Sensitivity analysis in observational research: introducing the E-value. *Ann Intern Med*. 2017;167:268–274. doi: 10.7326/M16-2607
20. Szklarczyk D, Kirsch R, Koutrouli M, Nastou K, Mehryar F, Hachilif R, Gable AL, Fang T, Doncheva NT, Pyysalo S, et al. The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Res*. 2023;51:D638–D646. doi: 10.1093/nar/gkac1000
21. Kanehisa M, Furumichi M, Tanabe M, Sato Y, Morishima K. KEGG. New perspectives on genomes, pathways, diseases and drugs. *Nucleic Acids Res*. 2017;45:D353–D361. doi: 10.1093/nar/gkw1092
22. Shi J, Guan J, Jiang B, Brenner DA, Del Monte F, Ward JE, Connors LH, Sawyer DB, Semigran MJ, Macgillivray TE, et al. Amyloidogenic light chains induce cardiomyocyte contractile dysfunction and apoptosis via a non-canonical p38alpha MAPK pathway. *Proc Natl Acad Sci U S A*. 2010;107:4188–4193. doi: 10.1073/pnas.0912263107
23. Ton VK, Mukherjee M, Judge DP. Transthyretin cardiac amyloidosis: pathogenesis, treatments, and emerging role in heart failure with preserved ejection fraction. *Clin Med Insights Cardiol*. 2014;8:39–44. doi: 10.4137/CMC.S15719
24. Park YH, Hodges A, Risacher SL, Lin K, Jang J-W, Ahn S, Kim SY, Lovestone S, Simmons A, Weiner MW, et al; AddNeuroMed consortium and the Alzheimer's Disease Neuroimaging Initiative. Dysregulated Fc gamma receptor-mediated phagocytosis pathway in Alzheimer's disease: network-based gene expression analysis. *Neurobiol Aging*. 2020;88:24–32. doi: 10.1016/j.neurobiolaging.2019.12.001
25. Heckmann BL, Teubner BJW, Tummers B, Boada-Romero E, Harris L, Yang M, Guy CS, Zakharenko SS, Green DR. LC3-associated endocytosis facilitates beta-amyloid clearance and mitigates neurodegeneration in murine Alzheimer's disease. *Cell*. 2019;178:536–551.e14. doi: 10.1016/j.cell.2019.05.056
26. Sorrentino V, Romani M, Mouchiroud L, Beck JS, Zhang H, D'Amico D, Moullan N, Potenza F, Schmid AW, Rietsch S, et al. Enhancing mitochondrial proteostasis reduces amyloid-beta proteotoxicity. *Nature*. 2017;552:187–193. doi: 10.1038/nature25143
27. Packer M. Critical reanalysis of the mechanisms underlying the cardiorenal benefits of SGLT2 inhibitors and reaffirmation of the nutrient deprivation signaling/autophagy hypothesis. *Circulation*. 2022;146:1383–1405. doi: 10.1161/CIRCULATIONAHA.122.061732
28. Porcari A, Cappelli F, Nitsche C, Tomasoni D, Sinigiani G, Longhi S, Bordignon L, Masri A, Serenelli M, Urey M, et al. SGLT2 inhibitor therapy in patients with transthyretin amyloid cardiomyopathy. *J Am Coll Cardiol*. 2024;83:2411–2422. doi: 10.1016/j.jacc.2024.03.429
29. Magaud C, Harnois T, Sebille S, Chatelier A, Faivre J-F, Bois P, Page G, Gellen B. Pro-inflammatory cytokine secretion induced by amyloid transthyretin in human cardiac fibroblasts. *Biochem Biophys Res Commun*. 2022;642:83–89. doi: 10.1016/j.bbrc.2022.12.027
30. Ioannou A, Patel RK, Ravi Y, Porcari A, Sinagra G, Venneri L, Bandera F, Masi A, Williams GE, O'Beara S, et al. Impact of earlier diagnosis in cardiac ATTR amyloidosis over the course of 20 years. *Circulation*. 2022;146:1657–1670. doi: 10.1161/CIRCULATIONAHA.122.060852
31. Chan N, Teruya S, Mirabal A, Weinsaft AY, Santos Jeffery Los DE, Guadalupe S, Jimenez M, Rodriguez C, Helmke S, Cuomo M, et al. Temporal outcomes of patients diagnosed with transthyretin cardiac amyloidosis. *J Card Fail*. 2024;30:1100–1107. doi: 10.1016/j.cardfail.2024.02.014
32. Candia J, Daya GN, Tanaka T, Ferrucci L, Walker KA. Assessment of variability in the plasma 7k SomaScan proteomics assay. *Sci Rep*. 2022;12:17147. doi: 10.1038/s41598-022-22116-0
33. Kraemer S, Vaught JD, Bock C, Gold L, Katilius E, Keeney TR, Kim N, Saccomano NA, Wilcox SK, Zichi D, et al. From SOMAmers-based biomarker discovery to diagnostic and clinical applications: a SOMAmers-based, streamlined multiplex proteomic assay. *PLoS One*. 2011;6:e26332. doi: 10.1371/journal.pone.0026332
34. Hensley P. SOMAmers and SOMAscan – A protein biomarker discovery platform for rapid analysis of sample collections from bench top to the clinic. *J Biomol Tech*. 2013;24:S5
35. Shimada YJ, Hasegawa K, Kochav SM, Mohajer P, Jung J, Maurer MS, Reilly MP, Fifer MA. Application of proteomics profiling for biomarker discovery in hypertrophic cardiomyopathy. *J Cardiovasc Transl Res*. 2019;12:569–579. doi: 10.1007/s12265-019-09896-z
36. Shimada YJ, Raita Y, Liang LW, Maurer MS, Hasegawa K, Fifer MA, Reilly MP. Comprehensive proteomics profiling reveals circulating biomarkers of hypertrophic cardiomyopathy. *Circ Heart Fail*. 2021;14:e007849. doi: 10.1161/CIRCHEARTFAILURE.120.007849
37. Lumish HS, Liang LW, Hasegawa K, Maurer MS, Fifer MA, Reilly MP, Shimada YJ. Prediction of worsening heart failure in hypertrophic cardiomyopathy using plasma proteomics. *Heart*. 2023;109:1837–1843. doi: 10.1136/heartjnl-2023-322644